

Supplementary Material: Exact Radial-Determinant Flexural Frequencies of a Monolithic Thickness-Poled Piezoelectric Ring: An Open-Circuit-Equals-Short-Circuit Theorem and a Segmented-Electrode Sensing Admittance

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Main text contains no job numbers; they are listed in §S.8.

S.1 Cubic consistency condition and the open-circuit-equals-short-circuit theorem

S.1.1 Short-circuit potential and the projected electrical equation

With both faces grounded, $\phi(\pm H) = 0$, write $\phi = \bar{\phi}(r, \theta, t)f(z)$ with $f(\pm H) = 0$, so $E_z = -\bar{\phi}f'$ and $E_{||} = -f\nabla\bar{\phi}$. With $\varepsilon_{rr} + \varepsilon_{\theta\theta} = -z\nabla^2 w$, the electric enthalpy $\mathcal{H} = \frac{1}{2}\varepsilon \cdot \bar{c}\varepsilon - \bar{e}_{31}E_z(\varepsilon_{rr} + \varepsilon_{\theta\theta}) - \frac{1}{2}\bar{\Xi}_{33}E_z^2 - \frac{1}{2}\bar{\Xi}_{11}|E_{||}|^2$ integrates through the thickness to

$$\mathcal{H}_2 = U_{\text{el}}(w) - \bar{e}_{31}I_1 \bar{\phi} \nabla^2 w - \frac{1}{2}\bar{\Xi}_{33}I_2 \bar{\phi}^2 - \frac{1}{2}\bar{\Xi}_{11}I_3 |\nabla\bar{\phi}|^2, \quad (\text{S.1})$$

with $I_1 = \int z f' dz$, $I_2 = \int f'^2 dz$, $I_3 = \int f^2 dz$ over $|z| \leq H$. Stationarity in w and in $\bar{\phi}$ (Hamilton's principle, the same weight for both fields) gives the flexural equation (A) below with $K_{\text{pref}} = -\bar{e}_{31}I_1$, the electrical equation $\bar{\Xi}_{11}I_3 \Delta\bar{\phi} - \bar{\Xi}_{33}I_2 \bar{\phi} + K_{\text{pref}}\Delta w = 0$, and the natural rim condition $\partial\bar{\phi}/\partial r = 0$. In the long-wave limit $\bar{\phi} \rightarrow (K_{\text{pref}}/\bar{\Xi}_{33}I_2)\Delta w$ and $D_{\text{SC}} - d = (\bar{e}_{31}I_1)^2/(\bar{\Xi}_{33}I_2)$. By Cauchy-Schwarz, $I_1^2 \leq I_2 \int z^2 dz = \frac{2}{3}H^3 I_2$, with equality only for $f' \propto z$: the quadratic $f = 1 - z^2/H^2$ is the unique one-term profile whose long-wave increment is the exact thin-plate value $\frac{2}{3}H^3 \bar{e}_{31}^2/\bar{\Xi}_{33}$ (from $\partial_z D_z = 0$ with $\phi(\pm H) = 0$). Its integrals are $I_1 = -4H/3$, $I_2 = 8/(3H)$, $I_3 = 16H/15$, so

$$M_{rr}^{\text{piezo}} = -\bar{e}_{31} \int_{-H}^H z E_z dz = -\frac{4H}{3} \bar{e}_{31} \bar{\phi}, \quad K_{\text{pref}} = \frac{4}{3} H \bar{e}_{31}. \quad (\text{S.2})$$

For comparison, the sinusoidal profile $f = \cos(\pi z/2H)$ gives $K_{\text{pref}} = (4H/\pi)\bar{e}_{31}$ and a long-wave increment $96/\pi^4$ of exact under this projection; closing its electrical equation instead by the thickness-integrated Gauss law (Duan, Quek and Wang's Eq. (12) [1]) replaces $\bar{\Xi}_{33}I_2\bar{\phi}$ by the face jump of D_z and gives an increment $12/\pi^2 = 1.216$ times exact. That closure was used in an earlier version of

this model; its free-free split (at $e_{31} = +4.1$, $+2.347\%$ against $+1.939\%$) is the 83% finite-element recovery discussed in §S.3. The full moment and shear are

$$M_{rr} = -\left[d\partial_r^2 w + (d - 2A_1)(r^{-1}\partial_r w + r^{-2}\partial_\theta^2 w) + K_{\text{pref}}\bar{\phi}\right], \quad Q_r = -\left[d\partial_r \nabla^2 w + K_{\text{pref}}\partial_r \bar{\phi}\right], \quad (\text{S.3})$$

with twisting moment $M_{r\theta} = -2A_1\partial_r(r^{-1}\partial_\theta w)$, which carries no piezoelectric term. The free-edge natural conditions are $M_{rr} = 0$ and the Kirchhoff effective shear $Q_r^{\text{eff}} = Q_r + r^{-1}\partial_\theta M_{r\theta} = 0$. On a Helmholtz branch $\Delta Z = \lambda Z$ at angular order n this is $Q_r^{\text{eff}}[cZ] = -[KZ' - 2A_1n^2(Z'/r^2 - Z/r^3)]$, with $K = d\lambda + K_{\text{pref}}\chi$. It reduces to Q_r at $n = 0$. The flexural motion equation is

$$d\Delta\Delta\bar{w} + K_{\text{pref}}\Delta\bar{\phi} - A_2\omega^2\bar{w} = 0. \quad (\text{A})$$

The projected electrical equation, divided by $\bar{\Xi}_{33}I_2$, is

$$a\Delta\bar{\phi} - \bar{\phi} + c\Delta\bar{w} = 0, \quad a = \frac{2H^2\bar{\Xi}_{11}}{5\bar{\Xi}_{33}}, \quad c = \frac{H^2\bar{e}_{31}}{2\bar{\Xi}_{33}}. \quad (\text{B})$$

Positing $\bar{\phi} = \chi\bar{w}$ with χ independent of r , (B) forces $\Delta\bar{w} = \lambda_B\bar{w}$ with $\lambda_B = \chi/(a\chi + c)$, and (A) forces $d\lambda^2 + K_{\text{pref}}\chi\lambda - A_2\omega^2 = 0$. Eliminating λ between the two gives a cubic in χ :

$$\begin{aligned} a_3\chi^3 + a_2\chi^2 + a_1\chi + a_0 &= 0, \\ a_3 &= K_{\text{pref}}a, \quad a_2 = d + K_{\text{pref}}c - A_2\omega^2a^2, \\ a_1 &= -2A_2\omega^2ac, \quad a_0 = -A_2\omega^2c^2. \end{aligned} \quad (\text{S.4})$$

Three roots $\chi_i(\omega)$ give three Helmholtz branches $\Delta\bar{w} = \lambda_i\bar{w}$, two Bessel functions per branch: six constants, matching the sixth-order coupled system. $\chi = 0$ is a root only at $\bar{e}_{31} = 0$ ($a_0 \propto \bar{e}_{31}^2\omega^2$), which is why the purely elastic ring is assembled as a separate 4×4 (§S.2) rather than by zeroing $\bar{e}_{31}$ inside this cubic. These coefficients are *not* obtained from the layered plate's own cubic [3] by the substitution $h_1 \rightarrow 2H$: the two potential shapes have opposite parity about their respective plate midplanes (this ansatz is even in z about the monolithic plate's own midplane; the layered plate's per-skin interior profile is neither even nor odd about the sandwich midplane), so the two cubics are genuinely different operators sharing only their general architecture.

The insulated-rim electrical condition, $\partial\bar{\phi}/\partial r = 0$ at r_i and r_o , together with two mechanical rows per edge, fills the 6×6 . Free-free uses $M_{rr} = Q_r^{\text{eff}} = 0$ at both edges; clamped-clamped uses $w = w' = 0$; clamped-free and free-clamped use one of each pairing at the two edges (§S.6).

S.1.2 Open-circuit and the coincidence theorem

With the bus ansatz $\phi = Vz/(2H) + \bar{\phi}(1 - z^2/H^2)$, the linear term's field $E_z^{\text{lin}} = -V/(2H)$ is spatially constant – even in z – so $\int_{-H}^H z\sigma^{\text{lin}}dz = 0$ identically: the linear piece never enters M_{rr} , for any V , H , or material constant. Because the bus is a single equipotential surface with no r, θ dependence, $E_{\parallel}^{\text{lin}} = 0$, and the enthalpy cross term $-\bar{\Xi}_{33}E_z^{\text{lin}}E_z^f$ integrates with weight $\int_{-H}^H f' dz = f(H) - f(-H) = 0$; so the linear piece drops from both (A) and (B): the open-circuit eigenvalue problem for $(w, \bar{\phi})$ is exactly the short-circuit pair, and V is fixed only afterward, by the scalar charge constraint $Q = 0$. The bus charge is the variable conjugate to V in the same enthalpy,

$$Q = -\frac{\partial}{\partial V} \int_A \mathcal{H}_2 dA = -C_0V, \quad C_0 = \frac{\bar{\Xi}_{33}A}{2H}, \quad (\text{S.5})$$

with no motional term: the strain coupling of the linear field carries the weight $\int_{-H}^H z dz = 0$, and $\bar{\phi}$ does not enter by the cross-term argument above. Hence $V = 0$ on an open-circuit bus for every flexural mode, and open-circuit coincides with short-circuit. (Integrating Gauss’s law pointwise to the face instead gives a charge with motional terms in $\int_A \nabla^2 w$ and $\int_A \bar{\phi}$; for the full face these cancel, the full-face $Q \equiv 0$ result of §S.7, so the two routes agree on the full electrode.) This is the theorem stated in the main text: because V never enters M_{rr} – not merely on a special mechanical edge, as on the layered plate, but identically – open-circuit flexure coincides with short-circuit flexure for every mechanical boundary condition, not only clamped–clamped. On the layered plate the coincidence is a corollary of $S = 0$ under clamped–clamped conditions specifically [3]; here it holds even when $S \neq 0$, because the moment channel for V does not exist at all on this stacking.

S.2 Elastic-limit gate

Setting $\bar{e}_{31} = 0$, $\bar{\phi} = 0$ in (A) recovers the isotropic Kirchhoff ring equation with $D = d = \frac{2}{3}\bar{c}_{11}^E H^3$, $\nu = \nu_p = \bar{c}_{12}^E/\bar{c}_{11}^E$, plate thickness $h = 2H$. Matching the standard isotropic bending stiffness $D = Eh^3/[12(1 - \nu^2)]$ requires $E = \bar{c}_{11}^E(1 - \nu_p^2)$. At the material point of the main text this gives $\bar{c}_{11}^E = 85.661$ GPa, $\bar{c}_{12}^E = 24.661$ GPa, $\nu_p = 0.28789$, $E = 78.561$ GPa. The free–free $n = 0$ fundamental of the new mixed-edge 4×4 at these constants agrees with an independently implemented closed-annulus Kirchhoff solver [2] at 4.77×10^{-10} relative. The coupled 6×6 is not exercised at this gate (§S.1: $\chi = 0$ is a root only at $\bar{e}_{31} = 0$, so the coupled operator is the wrong tool at exact zero coupling). This gate passed before any coupled number in the main text was trusted, and uses no piezoelectric code path.

S.3 Finite-element cross-check

Axisymmetric coupled-field finite elements (piezoelectric-plane elements, KEYOPT(1)=1001, Lanczos eigensolver), millimetre–newton– second–tonne units with relative permittivity, one PZT-4 rectangle $|z| \leq H = 10$ mm (no steel host), cloned from the mesh, material block, and electrical-boundary conventions already confirmed for the layered plate’s own free–free family [3], with the host volume deleted and the piezo volume filling the full thickness. Three decks on the same mesh: coupling-off (e0), both-faces short-circuit (scboth, VOLT= 0 at $z = \pm H$), and open-circuit (oc, VOLT= 0 at neither face, a single coupled VOLT set on both faces together, unconstrained). The coupled decks at $e_{31} = -4.1$ ran in job **2532086** (ANSYS 2025 R1); the coupling-off deck, which carries no e_{ij} , is from job **2520476**, where the same coupled decks ran at $e_{31} = +4.1$. Every deck ends RUN COMPLETED with **0 errors**, scored from the raw transcript and mode/discriminator files.

Mode 1 is the free–free rigid body (0 Hz) on every deck. Mode 2 is the first bending mode, identified by the through-thickness discriminator at $r = r_o$: transverse displacement same sign at $z = \pm H$, in-plane displacement opposite sign, ratio 19.7 (e0) and 20.4 (scboth, oc).

Table S.1: F–F $n = 0$ finite-element mode 2 (bending), $e_{31} = -4.1$.

deck	frequency [Hz]	closed form [Hz]	rel. vs closed form
e0 (coupling off)	73.506288	73.593	−0.118%
scboth (SC)	82.640099	82.735	−0.115%
oc (OC)	82.640099	82.735	−0.115%

scboth and oc mode 2 are bit-identical at the printed eight significant figures, 82.640099 Hz on both – the coincidence theorem confirmed in two-dimensional coupled-field finite elements, not only

in the Kirchhoff 6×6 – and so are the next several flexural overtones on the same pair of decks. **scboth** sits 12.426% above **e0**, against a closed-form +12.422% (ratio 1.0003); the finite-element short-circuit frequency sits -0.115% from the closed form, the same offset as the coupling-off deck (-0.118%), so the residual is the elastic Kirchhoff-versus-2D difference, not the coupling. At $e_{31} = +4.1$ (job 2520476) the same decks gave 74.933783 Hz on both, $+1.942\%$ against a closed-form $+1.939\%$; the finite elements follow the factor 6.4 between the two signs. **History.** The first version of this model closed the electrical equation by the thickness-integrated Gauss law with a sinusoidal profile (§S.1) and predicted $+2.347\%$ at $+4.1$; finite elements recovered 83% of it. The same closure predicted $+0.961\%$ for a clamped PZT-4 annular sector ($r_i/r_o = 0.03/0.07$, $2H = 0.8$ mm, $2\Theta = \pi$), where three-dimensional SOLID226 finite elements give $+0.782\%$ (ratio 0.814); the consistent projection gives $+0.796\%$ for that sector. Both ratios are $\pi^2/12 = 0.822$ to within the elastic offsets, which is how the projection was identified. Neither comparison was used to set a coefficient: the quadratic and its projection are fixed by the derivation of §S.1.

A companion extensional mode on the same mesh (mode 6 on **e0**, mode 5 on **scboth**) is 1669.57 Hz on both, and 1852.33 Hz on **oc** ($+10.95\%$; $+1.80\%$ at $e_{31} = +4.1$): the linear (membrane) piezoelectric field does couple to extension, confirming the negative control of main-text §2.1 in finite elements as well as in closed form. This is disclosed as claim C of this series [3] and is not built further here.

S.4 e_{31} inverse: gates, sensitivity, reflection twin

With $\bar{c}_{11}^E$ held at its known value, e_{31} is inverted from the free-free (**coupled_bisect**) and clamped-clamped (**cc_coupled_bisect**) short-circuit fundamentals of Table 1 of the main text ($e_{31} = -4.1$ rerun of the probe first archived as job **2520482**; **dps**=60, **inner_iters**= 45, **outer_iters**= 30; 727 s).

G_worker. The worker-process scalar reconstruction reproduces the in-process bisection roots at **rel**= 0.0 on both legs, before any inversion built on the in-process class is trusted.

G0 (round trip). Bracket $[-8.0, -1.0]$ C m $^{-2}$ on the physical branch (see the reflection-twin derivation below for why this range). Recovered $\hat{e}_{31} = -4.100000001$ against true -4.1 , **rel_err** = 2.499×10^{-10} (bar 10^{-6}), identical on both legs.

G1 (negative controls). A wrong-sign bracket $[1.0, 8.0]$ shows no sign flip; $e_{31} = +4.1$ gives a frequency $> 5\%$ from target on both legs (471.365 against 519.839, and 1757.694 against 1938.861 rad s $^{-1}$); $e_{31} = 0$ raises the singular-cubic exception of §S.1 when routed through a real determinant call (not at construction alone).

G2 (sensitivity). F-F: $d\omega/de_{31} = -7.661431$ rad s $^{-1}$ per C m $^{-2}$; propagated uncertainty 0.1655% of e_{31} at 0.01% frequency precision, 1.6549% at 0.1%. C-C: -30.70810 rad s $^{-1}$ per C m $^{-2}$; 0.1540% and 1.5400%. (At $e_{31} = +4.1$ the same legs gave 0.3178% and 0.3405%; the Gauss-law closure of §S.1 gave 0.2649% and 0.2822% there, its larger split making e_{31} look better conditioned. Fitted to a measured or finite-element split, that closure would return $|\bar{e}_{31}|$ low by $(\pi^2/12)^{1/2} = 0.907$.)

Reflection twin, derived. Let $\bar{e}_{31} = e_{31} - (c_{13}^E/c_{33}^E)e_{33}$. The cubic coefficients of (S.4) are odd in $\bar{e}_{31}$ for a_3, a_1 and even for a_2, a_0 (K_{pref} and c are odd, a is even). Flipping $\bar{e}_{31} \rightarrow -\bar{e}_{31}$ therefore maps the root set $\{\chi_i\} \rightarrow \{-\chi_i\}$ while each λ_i is invariant (numerator and denominator of $\lambda_B(\chi)$ flip together) and $K_{\text{pref}}\chi$ is invariant termwise (K_{pref} and χ each flip, the product does not): the determinant's M -row and Q -row blocks are unchanged, and only the $\bar{\phi}'$ -row block flips sign as a whole – an overall row rescaling that does not move where the determinant vanishes. Hence $e_{31}^{\text{twin}} = 2(c_{13}^E/c_{33}^E)e_{33} - e_{31}$ gives a bit-identical root frequency to e_{31} itself. At $e_{31} = -4.1$, $e_{31}^{\text{twin}} = 22.000869565217392$, confirmed to **rel**= 0 on both legs. The inversion bracket $[-8.0, -1.0]$ stays on

the printed branch, well away from $\bar{e}_{31} = 0$ ($e_{31} \approx 8.9504 \text{ C m}^{-2}$, where the cubic degenerates in degree) and from the twin. Do not chase the twin as a second physical solution; it is the same root, not a different one.

S.5 Overtones and thickness sweep

A sign-flip scan (dps= 60/inner_iters= 45, production settings; $e_{31} = -4.1$ rerun of the probe of job **2521175**) found three F–F pairs in $[455, 6000] \text{ rad s}^{-1}$ and two C–C pairs in $[1700, 6000]$; every pair reproduces the reflection twin of §S.4 at $\sim 5\text{--}6 \times 10^{-16}$ relative, not only at the fundamental.

Table S.2: F–F and C–C $n = 0$ overtones. Elastic and short-circuit (= open-circuit) roots in rad s^{-1} .

edge	mode	elastic	SC = OC	split
F–F	fundamental	462.398	519.839	+12.422%
F–F	OT1	2169.893	2398.891	+10.553%
F–F	OT2	5270.649	5850.693	+11.005%
C–C	fundamental	1726.890	1938.861	+12.275%
C–C	OT1	4776.965	5358.664	+12.177%

Short-circuit sits above elastic at every pair, both legs – the main-text stiffening-sign theorem generalizes beyond the fundamental. The split trend is not monotonic: F–F dips from 12.42% to 10.55% at the first overtone, then ticks back up to 11.01% at the second; C–C stays flatter, 12.27% $\rightarrow$ 12.18%. The on-branch sensitivity gives 0.166%, 0.181% and 0.171% of e_{31} at 0.01% frequency precision on the three F–F pairs.

A nine-point thickness sweep (F–F $n = 0$ only, fixed r_i, r_o ; $e_{31} = -4.1$ rerun of the probe of job **2521176**) tests the long-wave prediction that the split is independent of H at leading order.

Table S.3: F–F $n = 0$ fundamental versus ceramic thickness, fixed radii. Long-wave closed-form estimate of the thin-limit split: 13.13%.

H/r_o	elastic [rad/s]	SC [rad/s]	split
0.005	138.719	155.958	12.4271%
0.010	277.439	311.912	12.4258%
0.01667	462.398	519.839	12.4225%
0.025	693.596	779.714	12.4160%
0.040	1109.754	1247.343	12.3981%
0.060	1664.632	1870.413	12.3620%
0.090	2496.947	2803.657	12.2834%
0.130	3606.702	4044.544	12.1397%
0.180	4993.895	5588.893	11.9145%

Every point returns exactly one elastic and one coupled root in its search window; short-circuit exceeds elastic at every point; the thinnest point’s split (12.427%) is within a factor of 0.946 of the raw long-wave estimate $\frac{1}{2}\bar{e}_{31}^2/(\bar{\Xi}_{33}\bar{c}_{11}^E) = 13.133\%$ (unrounded constants). At this coupling the linearized estimate overshoots: the frequency grows as the square root of the stiffened bending stiffness, $\sqrt{1 + \bar{e}_{31}^2/(\bar{\Xi}_{33}\bar{c}_{11}^E)} - 1 = 12.37\%$, within 0.5% of the thinnest point. The estimate also omits the stiffening of $\bar{c}_{12}^E$ by the same $\bar{e}_{31}^2/\bar{\Xi}_{33}$, which changes the effective Poisson ratio and enters through the free-edge rows; it is not attempted here. The H -independence prediction holds well below $H/r_o \approx 0.025$ (split flat near 12.42%) and then drifts down to 11.915% by $H/r_o = 0.18$, a

−4.13% relative drift from the thinnest point – a real, small finite-thickness correction beyond the long-wave leading order, consistent with Kirchhoff kinematics being best justified in exactly the thin regime the main-text geometry ($H/r_o \approx 0.0167$) already sits in.

S.6 Mixed-edge (C–F/F–C) bit-identical reduction proof

Clamped–free and free–clamped determinants are built from a single mixed-edge row-selection routine that picks, at each edge independently, either the (w, w') pair (clamped) or the $(M_{rr}, Q_r^{\text{eff}})$ pair (free), following the pattern already used for the elastic ring’s own mixed-edge family [2], adapted to the three-branch cubic of §S.1. Because free–free and clamped–clamped are literal special cases of this routine (both edges set to the same label), the reduction was checked bit-for-bit (`mpmath` value equality, not a tolerance), not only re-derived: at $n = 0$, $\omega = 500$ and $n = 2$, $\omega = 1500$, the mixed-edge routine with both edges ‘F’ (respectively ‘C’) reproduces the original free–free (respectively clamped–clamped) elastic and coupled determinants exactly, eight of eight comparisons. This is a stronger correctness guarantee than an independent derivation would give: a defect in the row-selection logic would almost certainly have broken the already-validated free–free and clamped–clamped paths too, not only left a silent error in the new clamped–free/free–clamped paths.

Physical anchors ($e_{31} = -4.1$ rerun of the probe of job **2521220**): clamped–free elastic= 1605.4427637454191, SC= 1827.039084868 (+13.803%); free–clamped elastic= 556.8844158380273, SC= 610.530334689 (+9.633%). Both stiffen, consistent with the main text’s stiffening-sign estimate (Eq. 7). The negative control ($e_{31} = 0$ refused) still fires through the new coupled methods. A worker-path regression – new scalar-reconstruction functions for each of the four new bisection methods – was confirmed bit-identical to in-process, both in a standalone check and in the local test suite, for all four new methods.

The genuine physical finding of main-text §4.2 – the inner edge dominating bending stiffness far more than the outer edge at this $r_i/r_o = 1/6$ – is reported as an empirical fact of this one geometry, with a plausible but unproven mechanism: at $n = 0$ the free-edge moment condition’s r -dependent term is $-(2A_1/r) Z'$, which is $r_o/r_i = 6\times$ more sensitive to a clamp override at r_i than at r_o , consistent with, but not a rigorous proof of, the observed asymmetry (clamped–free at 90.4% of the way from free–free to clamped–clamped; free–clamped at 7.5%).

S.7 Force-driven sensing admittance: construction and gates

The two-region twelve-unknown system of the main text’s numerical-method section is built from the same three χ -cubic branches as §S.1, split at the load radius r_F , with column order (region I then region II, three branches, two Bessel functions each) matching the free-vibration determinant’s own inner loop. On a Helmholtz function $\Delta Z = \lambda Z$, the physical $n = 0$ resultants collapse to $M_{rr}[cZ] = -(KZ - 2A_1 Z'/r)$, $Q_r[cZ] = -KZ'$, $K = d\lambda + K_{\text{pref}}\chi$. Six rim rows (three per edge, F–F SC) plus six interface rows – $[w] = [w'] = [M_{rr}] = [\bar{\phi}] = [\bar{\phi}'] = 0$ and the one inhomogeneous shear jump $[Q_r] = -F/(2\pi r_F)$ – close the twelve-unknown system, solved by an equilibrated linear solve (row and column scaling, tracked and un-scaled after solving) rather than a determinant search, because the third (evanescent) χ -branch produces radial arguments large enough that raw entries span more than fifty orders of magnitude at this geometry.

Quantity and sign in G1–G_worker. The solver’s `Q_segment/Y_sense` return, per unit force, $+ \frac{4\pi}{3} H \Xi_{11} r_* \bar{\phi}'(r_*)$. That is $-Q_{\text{in}}^{(0)}$ in the main-text notation: the dielectric-flux part only, with the

sign of flux into the electrode. The gate values below are quoted as the code returns them. The free charge is $Q_{\text{in}} = \kappa Q_{\text{in}}^{(0)}$ (§S.7.1).

Five gates, from the $e_{31} = -4.1$ rerun of the probe first archived as job **2521701** (that run, at +4.1, was bit-identical between sandbox and cluster):

- **G1, force-balance identity** (the load-bearing sign gate on the shear-jump right-hand side): $2\pi \int_{r_i}^{r_o} r w(r) dr = -F/(A_2 \omega^2)$ at $\text{rel_err} = 0$ at the working precision (**dps=60**), confirming the jump's sign without assuming it.
- **G2, pole location** (unique-continuation gate): the driven system's own homogeneous determinant and the free-vibration short-circuit bisection agree at the F–F fundamental, $519.8388116516741 \text{ rad s}^{-1}$, to $\text{rel_err} = 1.8 \times 10^{-13}$, the bisection tolerance.
- **G3, $\bar{e}_{31} \rightarrow 0$ limit**, scaled correctly (raw e_{31} and e_{33} together, so $\bar{e}_{31}$ itself scales to zero – see the note below): Q_{in} at scale $1, 10^{-1}, \dots, 10^{-4}$ is $1.151 \times 10^{-11}, 1.779 \times 10^{-12}, 1.789 \times 10^{-13}, 1.789 \times 10^{-14}, 1.789 \times 10^{-15}$; decade ratios 6.467, 9.947, 9.999, 10.000; log–log slope over the three deep decades is 0.999988, clean linear vanishing.
- **G_mu** (diagnostic): the two mechanical Helmholtz branches match the purely elastic wavenumber $\pm\mu = \pm 10.250145149520318$ at $\text{rel_err} = 0.0$ at scale 10^{-8} , and the third branch converges to the pure-electrical Helmholtz eigenvalue $1/a = 5\Xi_{33}/(2H^2\Xi_{11}) = 20497.61$ of (B) (with $\bar{\Xi}_{33} \rightarrow \Xi_{33}$, because e_{33} is scaled to zero together with e_{31}).
- **G_worker**: the worker-process point evaluation reproduces Y_{sense} bit-identically (real and imaginary parts) both at $\omega = 200$ ($Y_{\text{sense}} = 1.1507340378860122 \times 10^{-11}$) and at $\omega = 521.5$, just above the pole ($Y_{\text{sense}} = 1.4054086233545436 \times 10^{-9}$).

Methodological note. An earlier pass at G3, scaling only the raw constructor keyword **e31** toward zero while leaving **e33** fixed, appeared (at $e_{31} = +4.1$) to show Q_{in} plateauing near 1.04×10^{-11} instead of vanishing, and the two mechanical branches plateauing near $\lambda \approx \pm 9.557$ instead of the correct $\pm 10.250145\dots$. Root cause: $\bar{e}_{31} = e_{31} - (c_{13}^E/c_{33}^E)e_{33}$ is dominated by the unscaled **e33** cross term at this material point, so with **e33** held fixed, $\bar{e}_{31}$ plateaus near -8.951 (its value at raw $e_{31} = 0$) and never actually approaches zero – the singular limit being probed was never reached. This was root-caused by direct numerical probe, consistent with this series' own methodology: precision was ruled out first (**dps** 50/100/150 gave bit-identical wrong answers under the buggy scaling, a real mathematical limit of the wrong quantity, not a conditioning artifact), which is what forced tracing the discrepancy to the test harness rather than the solver. A permanent regression test pins the correct behavior down going forward.

Reported, not gated: sweeping the code quantity across the F–F short-circuit fundamental shows the resonance of main-text Figure 3 – magnitude climbing from $\sim 2 \times 10^{-10}$ well off-resonance to $\pm 2.30 \times 10^{-7}$ within 0.01 rad s^{-1} of 519.8388 , with a clean sign flip through the pole, textbook driven-response behavior for a transfer function with a simple pole at the mechanical resonance and no damping in the operator. With $\kappa = -1.961$ the plotted free charge is 1.961 times these values, reaching $\pm 4.51 \times 10^{-7} \text{ C N}^{-1}$ at the same two points.

S.7.1 Transverse-shear (e_{15}) contribution and finite-element check

Charge on a face segment. Take the column $r_i < r < r_*$, $|z| \leq H$. It holds no free charge, and its inner rim is insulated. The free surface charge on the electrodes is $\sigma_{\text{top}} = -D_z(H)$ and

$\sigma_{\text{bot}} = D_z(-H)$. Gauss's law then gives $Q_{\text{top}} + Q_{\text{bot}} = 2\pi r_* \int_{-H}^H D_r dz$. D_z is odd in z for flexure, so $Q_{\text{top}} = Q_{\text{bot}}$, and each is $\pi r_* \int D_r dz$. This step is exact.

Leading order. For $6mm$ symmetry poled along z , $T_{rz} = c_{44}^E \gamma_{rz} - e_{15} E_r$ and $D_r = e_{15} \gamma_{rz} + \Xi_{11} E_r$. In a thin plate T_{rz} is fixed by equilibrium: $\int T_{rz} dz = Q_r$, the same shear resultant as in the plate equation, piezoelectric moment included. Kirchhoff's $\gamma_{rz} = 0$ is a kinematic constraint, not a constitutive fact. Eliminating γ_{rz} gives $D_r = (e_{15}/c_{44}^E) T_{rz} + (\Xi_{11} + e_{15}^2/c_{44}^E) E_r$. With $E_r = -\bar{\phi}'(1 - z^2/H^2)$ this gives the main-text expression for $\int D_r dz$. The order counting, with k the radial wavenumber, is:

- $\bar{\phi} \sim \bar{e}_{31} H^2 k^2 w / \bar{\Xi}_{33}$, so the dielectric term is $\sim \Xi_{11} \bar{e}_{31} H^3 k^3 w / \bar{\Xi}_{33}$;
- $Q_r \sim \bar{c}_{11}^E H^3 k^3 w$, so the shear term is $\sim e_{15} \bar{c}_{11}^E H^3 k^3 w / c_{44}^E$.

Both are $O(H^3 k^3 w)$. The feedback of e_{15} into the $\bar{\phi}$ equation is $O((Hk)^2)$ smaller and is dropped. The leading-order D_z is identically zero, so nothing else enters.

Closed form. At leading order $\bar{\phi} = c \nabla^2 w$ with $c = H^2 \bar{e}_{31} / (2 \bar{\Xi}_{33})$, and $Q_r = -D_{\text{eff}} (\nabla^2 w)'$ with $D_{\text{eff}} = \frac{2}{3} H^3 (\bar{c}_{11}^E + \bar{e}_{31}^2 / \bar{\Xi}_{33})$. The ratio of the two terms is therefore a constant, which gives the main-text κ . Table S.4 evaluates it.

Table S.4: $\kappa = Q_{\text{in}}/Q_{\text{in}}^{(0)}$ from the main-text closed form. The paper uses $e_{31} = -4.1 \text{ C m}^{-2}$ as tabulated and $c_{44}^E = 26 \text{ GPa}$; the other entries are sensitivities.

	$\bar{e}_{31}$	$\kappa, c_{44}^E = 73 \text{ GPa}$	$\kappa, c_{44}^E = 26 \text{ GPa}$
$e_{31} = -4.1$ (this paper, as tabulated)	-13.050	-0.0547	-1.9612
$e_{31} = +4.1$ (sign control)	-4.850	-1.5851	-6.2581

Gates (probe `probe_piezo_p5_e15_segment_charge_2026-09-24.py` at $e_{31} = -4.1$, SENTINEL PASS_ALL):

- **G_A:** the code's shear bracket equals $-Q_r$ from the equilibrium integral $Q_r(r_*) = -(A_2 \omega^2 / r_*) \int_{r_i}^{r_*} w r dr$, to 10^{-40} .
- **G_B:** $Q_{\text{in}}^{(0)}$ equals minus the code value.
- **G_C:** κ from the driven twelve-unknown solve at $\omega = 100, 200, 800, 1000, 1200 \text{ rad s}^{-1}$ is within 0.2% of the closed form at 26 GPa and within 0.0014 of it at 73 GPa, where κ is small.

Finite elements. The model is axisymmetric PLANE223 with KEYOPT(1)=1001 [4]. It uses the material block and mesh family of §S.3 with c_{44}^E as a parameter, and a harmonic, undamped solve.

- **Loading.** A 1 N ring load acts at $r_F = 350 \text{ mm}$, spread over the through-thickness node line (it is symmetric in z , so the response is flexure only). Both faces are at 0 V.
- **Charge.** The charge on each inner face segment is the sum of nodal charge reactions for $r \leq r_*$. The corner node at r_* is split half and half; the inclusive and exclusive sums bracket the result by $\pm 0.4\%$ at the finest mesh.
- **Sign calibration.** A first load step on each mesh puts 1 V on the top inner segment with no force. The reported reaction is negative, $|C| = 1.35\text{--}1.45 \times 10^{-7} \text{ F}$, and the grounded electrodes carry the equal and opposite charge (total $< 10^{-16}$ of it). The nodal charge reaction of this

release is therefore minus the electrode charge, and that sign is applied. An earlier driven-voltage deck of this series already showed the same sign, with a negative capacitive reaction far from resonance.

Six decks were run at $e_{31} = -4.1$ (job 2532086): $c_{44}^E \in \{73, 26\}$ GPa and $(N_r, N_z) \in \{(40, 6), (80, 12), (160, 12)\}$. Bars are relative to $\max(|\kappa|, 1)$, fixed before the run because κ is small at 73 GPa. All gates pass:

- every deck completed with no errors;
- the top and bottom segment charges agree to $\leq 2 \times 10^{-6}$ relative;
- the full-face sum is $\leq 3 \times 10^{-6}$ of the segment charge;
- the two finest meshes agree to $< 0.01\%$;
- imaginary parts are zero.

Table S.5 gives the results.

Table S.5: Finite-element segment charge over $Q_{\text{in}}^{(0)}$ (mesh 160×12) against κ from the twelve-unknown solve at the same ω , $e_{31} = -4.1$. The pre-registered bar was 5% of $\max(|\kappa|, 1)$.

ω (rad s ⁻¹)	$c_{44}^E = 73$ GPa			$c_{44}^E = 26$ GPa		
	FE	κ	ratio	FE	κ	ratio
100	-0.05491	-0.05493	0.9997	-1.95934	-1.96191	0.9987
200	-0.05503	-0.05482	1.0038	-1.95912	-1.96161	0.9987
800	-0.05393	-0.05579	0.9667	-1.95536	-1.96432	0.9954
1000	-0.05393	-0.05586	0.9654	-1.95784	-1.96454	0.9966
1200	-0.05384	-0.05602	0.9612	-1.95985	-1.96498	0.9974

At 26 GPa the residual of 0.1–0.5% is the size expected from the $O((Hk)^2)$ thickness terms the thin-plate expansion drops ($Hk \approx 0.03$ – 0.1 here). At 73 GPa the shear and dielectric terms nearly cancel; the same absolute residual, at most 0.0022 or 0.2% of the dielectric-flux value, is up to 4% of κ . The dielectric-flux value alone predicts a ratio of +1 at both stiffnesses. The finite elements reject it in sign, in magnitude, and in the c_{44}^E dependence. The same six decks at $e_{31} = +4.1$ (job 2531984; $\kappa = -1.585$ and -6.258) gave ratios 0.996–0.999, so the finite elements also follow the change of κ with the sign of e_{31} .

S.7.2 Clamped and mixed-edge sensing

Construction. The twelve-unknown system takes its rim rows from the same C/F selection as the mixed-edge determinants of §S.6: a clamped rim imposes $w = w' = \bar{\phi}' = 0$, a free rim $M_{rr} = Q_r = \bar{\phi}' = 0$; interface and load rows are unchanged. The Kirchhoff segment charge reads only $\bar{\phi}'$ at the cut and is unchanged too.

Gates (probe `probe_piezo_p5_mixed_sense_2026-09-24.py`, $e_{31} = -4.1$, `dps=40`, SENTINEL PASS_ALL on both halves of the run):

- **G_reduce:** the new builder’s free–free path equals the free–free solver’s Y_{sense} to 3×10^{-41} .
- **G_kernel:** its determinant changes sign at the short-circuit roots of main-text Table 1, 1938.860873289 (C–C), 1827.039084868 (C–F) and 610.530334689 (F–C), to 10^{-15} relative.

- **G_force:** with supports, identity G1 of §S.7 gains the rim reactions, $2\pi(r_o q(r_o) - r_i q(r_i)) = 2\pi A_2 \omega^2 \int r w dr + F$ with $q = -Q_r$; it holds to $\leq 6 \times 10^{-41}$ at all twenty test points, with $\int r w dr$ in closed form (cross-checked once against quadrature).
- **G_recip:** Maxwell–Betti reciprocity between ring loads at 0.25 and 0.45 m holds to $\leq 10^{-40}$.
- **G_kappa:** the driven κ at $r_* = 0.30$ m is within 0.5% of the closed form at every test frequency at 26 GPa and within 0.004 of it at 73 GPa.

The rim row. With $e_{15}\gamma_{rz}$ kept, the insulated rim is $\int D_r dz = (e_{15}/c_{44}^E)Q_r - \frac{4H}{3}(\Xi_{11} + e_{15}^2/c_{44}^E)\bar{\phi}' = 0$. At a free rim $Q_r = 0$ ($n = 0$) and this is the solver’s $\bar{\phi}' = 0$ (M1). At a clamped rim it is not; the rim-corrected alternative (M2) keeps the Q_r term and depends on c_{44}^E . Neither is strictly valid inside the electrical boundary layer, about H wide, that the clamp excites. At 26 GPa the model gives $\kappa = -2.39$ (M1) against -1.55 (M2) at a cut $1.25H$ from a clamped inner rim, and -2.00 against -1.91 at $2.5H$; interior cuts are unaffected.

Finite elements (job 2532086, 24 decks, ANSYS 2025 R1): the four pairings, $c_{44}^E \in \{73, 26\}$ GPa, meshes 80×12 , 160×12 , 320×24 ; same element family, material block, geometry, ring load and load-step-1 sign calibration as §S.7.1. A clamp sets $U_X = U_Y = 0$ on every node of the rim line; rims are electrically insulated. Five frequencies per pairing at 0.2–0.8 of its fundamental (0.85 for free-free, to stay clear of a charge zero), each at least 8% from any zero of the Kirchhoff charge. Criteria were fixed before the run (**Project Knowledge/PAPER5_MIXED_EDGE_SENSING_2026-09-24.md**): every deck RUN COMPLETED with zero errors; one sign of the charge reaction; $|C_{\text{seg}}|$ in 5×10^{-8} – 3×10^{-7} F (observed 1.28 – 1.49×10^{-7}); top and bottom equal and the full face at most 10^{-5} of the segment charge; the two finest meshes within 1% at the interior cuts. All pass. At the interior cuts $r_* = 0.30$ and 0.45 m every (pairing, c_{44}^E) cell is within 5% of $\max(|\kappa|, 1)$ at every frequency (**CORRECTED**), and the free-free cells reproduce §S.7.1. At 26 GPa the finest-mesh ratio to κ is 0.986–1.017 at the four lower frequencies and 1.004–1.039 at the highest, largest for C–C at 0.45 m.

Near-rim cuts (descriptive, labelled at 3%): free-rim cuts read $M1 = M2$ as they must, and match at 26 GPa except the free-free cut $1.25H$ from the inner rim, which reads 0.98–1.04. At clamped-rim cuts the finite elements lie closer to M1 than to M2 in every case, one unconverged cut included (for example C–C, 26 GPa, $1.25H$: 0.91–0.94 of the M1 value against 1.40–1.45 of M2), but only two of the twelve clamped-rim (cut, c_{44}^E) cells meet M1 at 3%, and no cut does so at both stiffnesses. By the pre-registered rule the paper gives no near-clamp value and places the partition more than about $3H$ from a clamped rim.

S.8 Cluster job provenance

All jobs on The Mill [5], solver snapshot 2026-07-10.s10. The monolithic piezoelectric path is a new module, unused by the elastic or layered-piezoelectric defaults. Jobs up to 2531984 used $e_{31} = +4.1 \text{ C m}^{-2}$; the paper’s numbers are from the $e_{31} = -4.1$ rerun described at the end of this section, and those jobs are the sign control.

- 2520476 — F–F finite-element queue (e0/scboth/ oc), 3/3 RUN COMPLETED, 0 errors, 0 warnings. Mode 2 bit-identical OC/SC at 74.933783 Hz.
- 2520482 — F–F/C–C e_{31} inverse (+4.1), SENTINEL PASS_ALL, both legs, $\hat{e}_{31}$ rel_err 2.499×10^{-10} .

- 2521175 — $n = 0$ overtones (F–F/C–C), SENTINEL PASS_ALL, bit-identical to the in-sandbox smoke-test digits.
- 2521176 — thickness sweep (F–F, 9 points), SENTINEL PASS_ALL, **G_found**, **G_theorem**, and **G_longwave** all pass.
- 2521220 — C–F/F–C mixed-edge, SENTINEL PASS_ALL, **G_reduce** bit-identical F–F/C–C reduction, worker-path green.
- 2521701 — force-driven sensing admittance, SENTINEL PASS_ALL, all 5 gates, bit-identical to the in-sandbox run.
- 2531984 — segmented-electrode harmonic finite elements (§S.7.1), 6 decks, RUN COMPLETED, 0 errors, 0 warnings, ANSYS 2025 R1. Scored CORRECTED (ratio to κ 0.996–0.999 at both c_{44}^E) after the charge-reaction sign calibration; the uncalibrated score is kept with the decks.
- 2532086 — $e_{31} = -4.1$ finite elements, ANSYS 2025 R1, all RUN COMPLETED, 0 errors: the free-free **scboth** and **oc** decks of §S.3; the six segment-charge decks of §S.7.1 (CORRECTED); and the 24 mixed-edge sensing decks of §S.7.2 (CORRECTED in all eight cells).

Consistent-projection rerun (2026-09-23). Jobs 2520482, 2521175, 2521176, 2521220 and 2521701 used the Gauss-law closure of §S.1. All five probes were re-run unchanged against the corrected solver (default **projection='consistent'**; the old closure is kept as **projection='duan'** and reproduces the old numbers bit-for-bit). Every gate passes (SENTINEL PASS_ALL on all five), and every number in this document and the main text is from that rerun. The finite-element job 2520476 does not depend on the analytic model.

$e_{31} = -4.1$ **rerun (2026-09-24).** The material bundle now carries $e_{31} = -4.1 \text{ C m}^{-2}$ as printed in Duan, Quek and Wang’s and Liu, Wang and Quek’s Table 1. The probes behind every number in the main text and this document were copied with $e_{31} = -4.1$ and run at their production precision on a two-core cloud workstation: mixed-edge frequencies, thickness sweep, overtones, sensing admittance, e_{31} inverse, segment charge and mixed-edge sensing. The inverse bracket moved to $[-8, -1] \text{ C m}^{-2}$ with the wrong-sign control at $[1, 8]$; the sensing sweep’s pre-registered window moved with the root. All end SENTINEL PASS_ALL. Every earlier cluster archival run of these probes reproduced its off-cluster run bit-for-bit, so the cluster runs were not repeated. The package test suite passes, including a test that pins the -4.1 free-free short-circuit root.

References

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- [4] ANSYS, Inc., *ANSYS Mechanical APDL*, Release 2025 R1, ANSYS, Inc., Canonsburg, PA, USA (2025). Used for finite-element validation only.
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